Work in progress: A Lean 4 Formalization of the Classical Continuation Chain for Periodic Navier–Stokes on $\mathbb{T}^3$: Energy Identity, H^3 Commutator Reduction and the BKM Route

Companion paper to `BenFrohman/periodic-ns-classical-continuation`

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Work in progress — draft, subject to change. DOI: 10.5281/zenodo.23842556. Companion reduction paper: 10.5281/zenodo.23842296.

How to cite / What is claimed. In Lean 4 with Mathlib, we give a machine-checked reduction of Clay Millennium Statement (B) (periodic Navier–Stokes on $\mathbb{T}^3$) to the single hypothesis `PeriodicBKMIntegrability`. We prove this hypothesis is equivalent to global existence of smooth solutions for every smooth periodic initial field. The BKM criterion and H^3 control are proved unconditionally, using only the standard axioms. `PeriodicBKMIntegrability` remains open. The equivalence of vorticity integrability with continuation of smooth solutions is classical (Beale–Kato–Majda, 1984); the novelty here is the complete, `sorry`-free formalization. Cite as: B. S. Frohman, reduction paper, DOI 10.5281/zenodo.23842296; main paper, DOI 10.5281/zenodo.23842556.

Draft — tracks build in progress. This document reports exactly what the Lean development proves at the stated commit. It does *not* solve the Clay Millennium problem, and does not prove global regularity of the 3D Navier–Stokes equations. The step `PeriodicBKMIntegrability` is the open Clay core and remains an explicit hypothesis.

Abstract

We describe a Lean 4/Mathlib formalization of the classical continuation steps for smooth divergence-free solutions of the incompressible Navier–Stokes equations on the unit torus $\mathbb{T}^3$, along the Statement (B) route of the companion repository `fefferman-clay-navier-stokes`. Proved with no hypothesis Props: local existence for $\nu \geq 0$, uniqueness via an L^2 energy-difference estimate, the Fourier coefficient ODE, coefficient decay, the H^s energy identity, Parseval's identity on the unit cube, and the Kato–Ponce toolkit at $s = 3$ (integration by parts, Leibniz rule, a Gagliardo–Nirenberg inequality, a bound on every lower-order term, and vanishing of the top-order term from $\operatorname{div} u = 0$). Also proved: the $s = 3$ physical-space commutator bound `periodicPhysicalCommutatorBound`. Also proved: the low-frequency vorticity bound `periodicLowFreqVorticityBound`, and hence, with no hypothesis Props, H^3 Beale–Kato–Majda control (`periodicBKMHsControl_three`) and the periodic BKM criterion (`periodicBKMCriterion`). Statement (B) is proven conditionally on `PeriodicBKMIntegrability`, which is open (`clay_B_exact_of_integrability`). Small

zero-mean H^1 data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`.

1 Conventions and naming

Every hypothesis is a named `Prop` passed as an explicit theorem argument; no `axiom`, `sorry` or `admit` appears. The axiom audit (`lakeenvleanAxiomAudit.lean`) reports only `propext`, `Classical.choice` and `Quot.sound`. Lean identifiers defined in this repository are *coined in this build* (marked [coined]), not names from the author's May/June 2026 manuscript. The identifiers `Clay_B_exact`, `PeriodicLocalExistence`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes` (pinned at `ebcf109`).

2 Relation to the May/June 2026 manuscript

The author's manuscript frames regularity through a tethered Lyapunov functional S_ε and a comparison ODE $y' = Cy^2 - \kappa'' y^3$, with transport cancellation, viscous dissipation, a Hölder–Sobolev gate and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). For *unmodified* Navier–Stokes ($\kappa = 0$) the cubic damping is absent, and the remaining gap is hypothesis (H5): a non-circular majorant for the vorticity. In this repository that gap appears exactly as `PeriodicBKMIntegrability`, i.e. $\int_0^T \|\omega\|_\infty \, dt < \infty$. Nothing here closes it. The transport cancellation of the manuscript corresponds to `top_term_zero` below.

3 Proved results (no hypothesis Props)

Theorem 1 (Local existence). *`periodicLocalExistence_of_nonneg`[coined]: for $\nu \geq 0$, `PeriodicLocalExistence` ν holds (Galerkin method at $s = 3$).*

Theorem 2 (Energy difference / uniqueness). *`periodicEnergyDifferenceEstimate`[coined]: for $\nu \geq 0$ and two classical solutions with equal data, $E(t) = \int_{[0,1]^3} |u - v|^2$ satisfies $E' \leq 2CE$. With Grönwall (`eq_zero_of_deriv_le_mul`[coined], `periodicVelocityUniqueness_of_energy`[coined]) this gives uniqueness.*

Theorem 3 (Fourier coefficient ODE, decay, H^s energy identity). *`periodicFourierCoeffODE`[coined], `periodicFourierCoeffDecay`[coined] and `periodicFourierHsEnergyIdentity`[coined] (file `CoeffODE.lean`) prove the Props `PeriodicFourierCoeffODE`, `PeriodicFourierCoeffDecay`, `PeriodicFourierHsEnergyIdentity` for every ν (and every s): each Fourier coefficient $\hat{u}_k(t)$ obeys the projected ODE, coefficients decay faster than any power, and*

$$\tfrac{d}{dt}\|u\|_{H^s}^2 = -2\nu \textstyle\sum_k (1 + |k|^2)^s \, 4\pi^2 |k|^2 |\hat{u}_k|^2 + \textit{(nonlinear rate)}.$$

The bridge is `periodicFourierHsEnergyIdentity_of_coeff`[coined] (`HsEnergyIdentity.lean`).

Theorem 4 (Parseval on the unit cube). *`hasSum_prod_cubeCoeff`[coined] (`Parseval.lean`): for continuous periodic f, g, $\sum_k \overline{\hat{f}_k} \hat{g}_k = \int_{[0,1]^3} \bar{f} g$, via `torusLift`[coined] and Mathlib's torus Parseval.*

Theorem 5 (Physical form of the H^3 rate). *`hasSum_rate_three`[coined]: the Fourier-side H^3 nonlinear rate is summable and equals $-2\sum_i \int (1 - \Delta/4\pi^2)^3 v_i \, ((v \cdot \nabla) v)_i$.*

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Theorem 6 (Kato–Ponce toolkit, $s = 3$; `KatoPonce.lean`)**.** *Proved: unit-cell integration by parts* `cellI_ibp`[coined]*,* `cellI_ibp'`[coined]*; Leibniz rule* `DL_mul`[coined]*; Gagliardo–Nirenberg on the cell* `gagliardoNirenberg_cell`[coined] *and* `cellI_sq_mul_abs_le`[coined]*; and*

- `lower_term_bound`[coined]*: every lower-order Leibniz term (first factor of order* ≥ 1*) satisfies* $|\int D^L v_i \cdot D^{p_1} v_l \, D^{p_2} \partial_l v_i| \leq 5 \|\nabla v\|_\infty \, Q(v)$*, with* $Q =$ `physQ`[coined]*;*
- `top_term_zero`[coined]*:* $\sum_{i,l} \int D^L v_i \, v_l \, D^L \partial_l v_i = 0$*, because* $\operatorname{div} v = 0$ *(transport cancellation).*

Theorem 7 (Continuation alternative)**.** `periodicBKMCriterion_of_hsControl`[coined]*: for* $\nu \geq 0$*,* $s > 5/2$*,* `PeriodicBKMHsControl` ν s *implies* `PeriodicBKMCriterion` ν *(the* H^s *blow-up alternative is proved).*

4 Conditional results

Named hypothesis 1. `PeriodicPhysicalCommutatorBoundC`[coined] (s=3): for smooth periodic divergence-free v, $-2\sum_i \int (1 - \Delta/4\pi^2)^3 v_i \, ((v \cdot \nabla)v)_i \leq C \|\nabla v\|_\infty \|v\|_{H^3}^2$. *Status:* **proved** in `CommutatorAssembly.lean` : `periodicPhysicalCommutatorBound`[coined] gives it with $C = 2 \cdot 64 \cdot 315 \cdot 64 \, (4\pi^2)^3$, from `lower_term_bound`, `top_term_zero`, `word_bound`[coined] and `physQ_le_hsSq`[coined]. No longer a hypothesis.

Named hypothesis 2. `PeriodicLowFreqVorticityBound`[coined]: $\exists K \geq 0$, $\forall R \geq 2$, $\|V_R \nabla v\|_\infty \leq K(1 + \log R)\|\omega\|_\infty$. *Status:* **proved** in `LowFreqAssembly.lean` (`periodicLowFreqVorticityBound`[coined]). The proof writes the Biot–Savart law in Riesz form, $\widehat{\partial_j v_i} = q_{j,i+2}\hat{\omega}_{i+1} - q_{j,i+1}\hat{\omega}_{i+2}$ with $q_{jb} = k_j k_b / |k|^2$. It splits q into a low piece and $S = \lceil \log_2 R \rceil + 3$ smooth dyadic blocks, each with a uniformly L^1-bounded kernel (`DyadicBlock.lean`). Every piece is followed by the uniformly L^1-bounded de la Vallée Poussin kernel (`VPKernel.lean`). No longer a hypothesis.

Named hypothesis 3. `PeriodicBKMIntegrability` ν (from `fefferman-clay-navier-stokes`): $\int_0^T \|\omega\|_\infty < \infty$ on every finite interval of existence. *Status:* **open**. This is the Clay core and the manuscript's (H5) gap.

Proposition 8. `periodicFourierCommutatorBound_of_physical`[coined]*:* `PeriodicPhysicalCommutatorBoundC` $\Rightarrow$ `PeriodicFourierCommutatorBound3C`*.* `periodicBKMHsControl_of_physical_lowFreq`[coined]*:* ($\exists C \geq 0$, `PeriodicPhysicalCommutatorBoundC`) *and* `PeriodicLowFreqVorticityBound` *give* `PeriodicBKMHsControl` ν *3. Combined with the proved bound:* `periodicBKMHsControl_three_of_lowFreq`[coined] *gives* `PeriodicBKMHsControl` ν *3 from* `PeriodicLowFreqVorticityBound` *alone.*

Theorem 9 (unconditional corollaries, `BKMFinal.lean`)**.** `periodicBKMHsControl_three`[coined]*:* `PeriodicBKMHsControl` ν *3 for every* ν*, with no hypotheses.* `periodicBKMCriterion`[coined]*:* `PeriodicBKMCriterion` ν *for every* ν*, with no hypotheses.*

Proposition 10 (Statement (B), conditional)**.** `clay_B_exact_of_hsControl`[coined]*: for* $s > 5/2$*,* `PeriodicBKMHsControl` ν s *and* `PeriodicBKMIntegrability` ν *give* `Clay_B_exact` ν*. Also* `clay_B_exact_of_corrected_route'`[coined] *and* `clay_B_exact_of_bkm`[coined]*. The final form,* `clay_B_exact_of_integrability`[coined]*, is conditional* only *on the open* `PeriodicBKMIntegrability` ν*.*

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Note on a failed route. A Wiener-algebra/Fourier commutator estimate was tried and abandoned: its low-frequency part grows like $R^{3/2}$, not $\log R$. The $s = 3$ physical-space estimate replaces it.

5 Attacking the open hypothesis (work in progress)

Files `IntegrabilityAttack.lean`, `SmallDataAttack.lean`, `TetherClaimAttack.lean`, `EnstrophyIdentity.lean`, `TetherKappaZero.lean` (commits `f981e71`, `052cef1`, `ea5133d`). All names are coined. **This does not prove `PeriodicBKMIntegrability`.**

- `integrableOn_vorticity_of_lt_tstar` (proved): if $T < T^*$, then $\int_0^T \|\omega\|_\infty < \infty$.
- `periodicBKMIntegrability_iff_globalLifespan` (proved): `PeriodicBKMIntegrability` ν holds iff `PeriodicGlobalLifespan` ν ($T^* = \infty$ for all smooth data). The open hypothesis is therefore exactly global existence.
- Route 2 (proved from named classical hypotheses, not formalized here):
  - `integrability_of_smallDataGlobal`, from `PeriodicSmallDataGlobal`;
  - `integrability_of_prodiSerrin`, from `PeriodicProdiSerrinCriterion` (endpoint $L_t^\infty L_x^\infty$) plus a velocity bound.
- Route 1 (the $\kappa = 0$ analogue of the tether), `periodicBKMIntegrability_of_vorticityAPriori` (proved): from the **open** `PeriodicVorticityAPrioriBound`. It breaks because with no damping term the enstrophy satisfies only $E' \le CE^3$ and the max-point bound gives $Y' \lesssim Y^2 \log(\cdot)$. Neither gives a uniform bound.
- `periodicEnstrophyIdentity` (`EnstrophyIdentity.lean`, proved, **no hypotheses**): the $\mathbb{T}^3$ enstrophy (H^1) energy identity, the $s = 1$ case of the H^s identity.
- Small zero-mean H^1 data (`EnstrophyIdentity.lean`): `riccati_barrier_Ioo` (proved) and `periodicTstar_eq_top_of_smallH1_of_rate`, proved from the unconditional identity plus the named classical hypotheses `PeriodicEnstrophyRateBound` (Ladyzhenskaya/Poincaré trilinear bound) and `PeriodicH1BlowupCriterion` (Leray), both **not formalized**. *Bug fix:* the earlier hypothesis `PeriodicEnstrophyRiccati` with $\alpha > 0$ (`SmallDataAttack.lean`, used by `periodicTstar_eq_top_of_smallH1`, `integrability_of_smallH1`) is false for small constant flows, since the mean mode is conserved; the zero-mean rate-bound version replaces it.
- Tether claim (`TetherClaimAttack.lean`): `cubic_riccati_barrier` (proved); `periodicVorticityAPrioriBound_of_tetherCubicRate`, `clay_B_exact_of_tetherCubicRate` proved from the **open** `PeriodicTetherCubicRate` (not derived from unmodified NS); `uniform_cubic_rate_false_of_scaling` (proved, abstract scaling step: a data-uniform cubic rate is contradictory).
- Direction route: `integrability_of_directionCriterion`, `periodicBKMIntegrability_of_direction`, proved from the classical `PeriodicDirectionCriterion` (Beirão da Veiga–Berselli / Constantin–Fefferman, not formalized) plus the **open** `PeriodicDirectionRegularity`. New: `periodicDirectionCriterion_of_growth` proves `PeriodicDirectionCriterion` from the unconditional enstrophy identity, the linear Grönwall lemma

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`exp_gronwall_Ioo`, and two named open hypotheses: `PeriodicDirectionEnstrophyGrowth` (coined; Constantin–Fefferman 1993 geometric depletion of the strain kernel plus Beirão da Veiga–Berselli 2002, constant-rate form, not formalized) and `PeriodicH1BlowupCriterion` (Leray 1934). `periodicBKMIntegrability_of_directionGrowth` still needs the open `PeriodicDirectionRegularity`.

- $\kappa = 0$ tether (`TetherKappaZero.lean`): `tetherForm_zero`, `tetherGeometricControl_zero`, `tetherImpliesDirection_zero_iff` (proved): at $\kappa = 0$ the tether control is vacuous, so "tether control ⇒ direction regularity" is *equivalent* to the open `PeriodicDirectionRegularity`; `periodicBKMIntegrability_of_tetherKappaZero` (conditional).

6 Status

- **Proved, unconditional:** local existence ($\nu \geq 0$), uniqueness, coefficient ODE and decay, H^s energy identity, Parseval, physical form of the H^3 rate, Kato–Ponce toolkit including `lower_term_bound` and `top_term_zero`, the $s = 3$ commutator bound `periodicPhysicalCommutatorBound`, the blow-up alternative, the low-frequency bound `periodicLowFreqVorticityBound`, H^3 BKM control `periodicBKMHsControl_three` and the BKM criterion `periodicBKMCriterion`, the equivalence `periodicBKMIntegrability_iff_globalLifespan`, and the H^1 enstrophy identity `periodicEnstrophyIdentity`.
- **Conditional:** Statement (B) is proven conditionally on `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). Small zero-mean H^1 data give $T^* = \infty$ conditionally on `PeriodicEnstrophyRateBound` and `PeriodicH1BlowupCriterion`; at $\kappa = 0$ the tether headline is equivalent to the open `PeriodicDirectionRegularity`.
- **Open:** `PeriodicBKMIntegrability` — the Clay Millennium core. Neither the Clay problem nor global regularity of 3D Navier–Stokes is claimed.

Reproducibility

`lakebuild`; `lakeenvleanAxiomAudit.lean` (0 `sorry`; only `propext`, `Classical.choice`, `Quot.sound`).

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